

External-Prime Accumulation and the p -Adic Newton Wall

A nonduplicative progress report on the Alaoglu–Erdős integer-exponent conjecture

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Abstract

Let

$$2^x \in \mathbb{Z}, \quad 3^x \in \mathbb{Z}.$$

The conjecture of Alaoglu and Erdős asserts that $x \in \mathbb{Z}$. The supplied unified report already develops a very large corpus around the exact solution monoid, transcendence reductions, determinant and interpolation methods, structural-prime residuals, p -adic valuation lattices, Sturmian and carry encodings, canonical products, and geometric Newton interpolation. This report deliberately does not reproduce that corpus. It opens a new local direction at a prime $p \geq 5$ dividing a hypothetical output.

The first result is a rigidity theorem for geometric sampling in a nonarchimedean field. If $|Q|_p < 1$ and an analytic germ F at zero satisfies $F(Q^k) = R^k$ for every sufficiently large k , then $F(z) = z^d$ and $R = Q^d$ for some $d \in \mathbb{N}_0$. A flexible two-row version says that $F(\alpha Q^k) = \lambda F(Q^k)$ on an accumulating geometric ray forces the Taylor support of F into the exact eigenspectrum $\{n : \alpha^n = \lambda\}$. In the Alaoglu–Erdős setting the edge pairs are $(\alpha, \lambda) = (2, 3)$ or $(3, 2)$, so no nonzero analytic, meromorphic, or finite-Puiseux germ can realize even one adjacent strip of the conditional integer grid.

The second result analyzes, at the same external prime, the geometric Newton coefficients already identified in the companion report. Their valuations are computed exactly. The leading coefficient of the unique degree- N interpolation remainder forces every rational polynomial interpolating the first $N+1$ native ray values to contain a coefficient with an intrinsic denominator of order

$$p^{\frac{1}{2}v_p(Q)N^2 + O(N)}.$$

On any off-ray unit multiple uQ^k , the Newton terms have valuations tending quadratically to $-\infty$; eventually the last term uniquely dominates every partial sum. Thus the canonical Newton completion converges geometrically over $\mathbb{C}$, truncates exactly on its native geometric ray in every completion, but diverges maximally over $\mathbb{Q}_p$ at the adjacent Alaoglu–Erdős row. This is the *p -adic Newton wall*.

The report then gives a Tate-algebra formulation of the edge operator $F(z) \mapsto F(\alpha z) - \lambda F(z)$, an exact finite-strip defect lower bound, a normal-family obstruction to uniformly overconvergent interpolants, a finite-rank q -difference research interface, a computational audit, and a Lean formalization plan. These are genuine new lemmas and closure criteria, not a proof of the conjecture. The remaining bridge is sharply identified: one must derive nonarchimedean regularity, overconvergence, or finite-rank q -difference structure from the global arithmetic data without assuming the desired power branch is already regular at the accumulation point.

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1 Mandate, status, and separation from the supplied report

1.1 The problem and the standard conditional notation

The problem is

Conjecture 1.1 (Alaoglu–Erdős). *For every real x ,*

$$2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}.$$

It originates in the study of highly composite and related numbers [1]. It is a special instance of the still-open Four Exponentials boundary; see, for example, Waldschmidt’s survey [3].

Assume throughout the mathematical core that a nonintegral solution exists, and put

$$M = 2^x \in \mathbb{N}, \quad A = 3^x \in \mathbb{N}, \quad \theta = \frac{\log 3}{\log 2}. \quad (1)$$

Then $A = M^\theta$ in the ordinary real sense. The companion report [2] establishes much more conditional structure. Only the small package explicitly restated in [Section 3](#) is imported here.

1.2 What is new here

The supplied report already contains the complex entire completion of the geometric Newton interpolants and exact formulas for their coefficients. What it does not contain is the external-prime accumulation analysis carried out below. The new material is:

- N1.** a geometric-ray classification theorem for analytic and meromorphic nonarchimedean germs;
- N2.** an edge-law spectral theorem from zeros accumulating at the origin;
- N3.** the corresponding exclusion of analytic, meromorphic, and finite-Puiseux realizations of an adjacent conditional Alaoglu–Erdős strip;
- N4.** exact external-prime valuations of every geometric Newton coefficient, including the resonant valuation-tie correction;
- N5.** a basis-independent quadratic lower bound for the p -adic Gauss norm of *every* polynomial interpolant through a native geometric prefix;
- N6.** exact off-ray valuations, an exact term-ratio formula, and eventual last-term dominance for Newton partial sums;
- N7.** a Tate-algebra spectral-gap estimate for the dilation edge operator and a finite-strip quotient lower bound;
- N8.** a compactness theorem showing that exact finite interpolants cannot have a uniformly overconvergent p -adic subsequence;
- N9.** a concrete Robba-to-Tate and finite-rank q -difference research interface;
- N10.** a reproducible exact-arithmetic audit and a staged Lean formalization plan.

Status convention

Proved here means that a complete conventional proof is included. **Imported** means that the statement is taken from the supplied unified report or a cited classical source and used only as an input. **Conditional target** means that the implication is proved but the additional hypothesis has not been derived from a hypothetical counterexample. **Audit/no-go** records a rigorously identified obstruction and is not presented as progress on the truth of the conjecture itself.

2 Public-state check as of 23 August 2026

The user’s competitive premise is substantially grounded, but two headlines need precise qualification.

1. Anthropic publicly launched Claude Fable 5 on 9 June 2026 and describes it as its most capable generally available model for long-horizon work [11].
2. The Jacobian development is an explicit Fable-assisted counterexample in dimension three, hence in every dimension at least three. The two-dimensional Jacobian conjecture remains open. Tao gives the explicit degree-seven map and a short verification [12]. Thus “resolved the Jacobian conjecture” is accurate only if it is understood as disproving the unrestricted conjecture, not settling the remaining planar case.
3. A curve with thirty exhibited independent rational points was posted to the ICARM leaderboard and attributed to Claude working with Levent Alpöge and Ava Howell. This proves an unconditional rank lower bound of 30; an unconditional proof that the exact rank equals 30 has not been claimed [13, 14].
4. The reported Hadamard construction supplies matrices for all twelve previously missing admissible orders below 2000, including order 668. Epoch AI records the result and its human/AI attribution provisionally; OEIS records the twelve orders and attribution [15, 16]. This closes that finite table, not the general Hadamard conjecture.

Exact web searches for “Alaoglu–Erdős”, the integer-exponent statement, Fable, and Lean found no public preprint, repository, announcement, or indexed social post describing the rumored new route as of the date of this report. I therefore treat the rival team’s message as unverified tactical information rather than as a mathematical input. The search below is independent of any claimed method.

3 Minimal counterexample package and the external orientation

3.1 Elementary consequences

Lemma 3.1. *Under (1), a nonintegral solution x is positive, irrational, and transcendental. Moreover, M is not a power of 2, A is not a power of 3, and 2, M are multiplicatively independent, as are 3, A .*

Proof. If $x < 0$, then $0 < 2^x < 1$, contradicting integrality; $x = 0$ is integral. If $x = a/b \in \mathbb{Q}$ in lowest terms with $b > 0$, then $2^a = M^b$. Comparing the exponent of the prime 2 gives $b \mid a$, hence $b = 1$. Thus a nonintegral solution is irrational. If it were algebraic irrational, Gelfond–Schneider

applied to $2^x = M \in \overline{\mathbb{Q}}$ would be a contradiction, so it is transcendental. The remaining statements follow immediately: for example $M = 2^m$ gives $x = m$, and $2^r M^s = 1$ gives $r + sx = 0$, forcing $r = s = 0$ because x is irrational. $\square$

The next fact is imported by the supplied corpus, but its short proof is included because it is the control-sensitive hinge of this report.

Lemma 3.2 (External-prime lemma). *If (1) holds with $x \notin \mathbb{Z}$, then some prime $p \geq 5$ divides MA .*

Proof. Suppose instead that both M and A are $\{2, 3\}$ -smooth. Write

$$M = 2^a 3^b, \quad A = 2^c 3^d \quad (a, b, c, d \in \mathbb{N}_0).$$

With $\theta = \log 3 / \log 2$, the two logarithmic identities give

$$x = a + b\theta, \quad x\theta = c + d\theta,$$

so

$$b\theta^2 + (a - d)\theta - c = 0.$$

The number θ is transcendental: it is not rational by unique factorization, and if it were algebraic irrational then $2^\theta = 3$ would contradict Gelfond–Schneider. Therefore $b = c = 0$ and $a = d$, whence $x = a \in \mathbb{Z}$, a contradiction. $\square$

3.2 Oriented external-prime notation

Choose an external prime $p \geq 5$ from Theorem 3.2. At least one of M, A is p -adically contracting. We orient the data as follows:

$$(Q, R; \alpha, \lambda, \gamma) = \begin{cases} (M, A; 2, 3, \theta), & p \mid M, \\ (A, M; 3, 2, \theta^{-1}), & p \mid A. \end{cases} \quad (2)$$

If p divides both, either orientation may be used. In both cases

$$p \mid Q, \quad Q^\gamma = R, \quad \alpha^\gamma = \lambda, \quad (Q^k)^\gamma = R^k, \quad (\alpha Q^k)^\gamma = \lambda R^k. \quad (3)$$

Set

$$m = v_p(Q) > 0, \quad a = v_p(R) \geq 0. \quad (4)$$

Because $p \geq 5$, all of $2, 3, \alpha, \lambda$, and $\alpha - 1$ are p -adic units. The points Q^k and αQ^k tend to 0 in $\mathbb{Q}_p$, giving a genuine nonarchimedean accumulation point that is absent from the ordinary real geometry.

The control-sensitive feature

For an integer control $x = d$, one has $(M, A) = (2^d, 3^d)$, so no prime $p \geq 5$ divides MA . The oriented external-prime package (2) therefore cannot even be formed for controls. This does not by itself prove the conjecture, but it passes the supplied report's control-principle test: every theorem below visibly consumes a fact unavailable to the integer controls.

4 Geometric rays at a nonarchimedean accumulation point

4.1 One-ray analytic classification

The following elementary theorem is the cleanest new structural result in this report.

Theorem 4.1 (Geometric-ray rigidity). *Let K be a complete nonarchimedean field of characteristic zero. Let $t, s \in K^\times$ with $0 < |t| < 1$, and let F be analytic on a neighborhood of 0. If*

$$F(t^k) = s^k \tag{5}$$

for every sufficiently large integer k , then there is a unique $d \in \mathbb{N}_0$ such that

$$s = t^d, \quad F(z) = z^d$$

on a neighborhood of 0.

Proof. The values in (5) are nonzero, so F is not identically zero. Write

$$F(z) = z^d G(z), \quad d = \text{ord}_0 F \in \mathbb{N}_0, \quad G(0) = c \neq 0.$$

Then

$$\left(\frac{s}{t^d}\right)^k = G(t^k) \longrightarrow c. \tag{6}$$

Put $u = s/t^d$. Since the limit is nonzero, $|u| = 1$. A convergent sequence u^k with $|u| = 1$ can converge to a nonzero limit only if $u = 1$: indeed

$$u^{k+1} - u^k = u^k(u - 1) \longrightarrow 0,$$

and $|u^k| = 1$ forces $u - 1 = 0$. Hence $s = t^d$ and $G(t^k) = 1$ for all sufficiently large k . The analytic function $G - 1$ has infinitely many zeros tending to the interior point 0, so the nonarchimedean identity theorem gives $G \equiv 1$. $\square$

Corollary 4.2 (One-ray obstruction for a hypothetical counterexample). *In the oriented setting (2), no analytic germ F at 0 can satisfy $F(Q^k) = R^k$ for every sufficiently large k .*

Proof. By Theorem 4.1, such a germ would give $R = Q^d$ for some $d \in \mathbb{N}_0$. In the first orientation this is $3^x = (2^x)^d$, hence $3 = 2^d$ because $x > 0$. In the second it is $2 = 3^d$. Both are impossible. $\square$

Remark 4.3 (Newton-polygon reading). The proof separates two obstructions. First, an analytic germ has an integral order d at zero, so valuation comparison along the ray forces

$$\frac{v_p(R)}{v_p(Q)} = \frac{a}{m} = d \in \mathbb{N}_0.$$

If a/m is not an integer, the Newton slope already rules out analyticity. If $a = dm$, the unit quotient R/Q^d must also have all of its powers converging; this forces the quotient to be exactly 1, not merely a root of unity or a principal unit. Thus the valuation slope is only the first layer; the unit mismatch is decisive in the resonant case.

4.2 The edge-law spectral theorem

The two-row formulation no longer assumes that the native values themselves form a geometric progression.

Theorem 4.4 (Accumulating edge law). *Let K be as above, let $0 < |t| < 1$, and let $q, \mu \in K^\times$. Suppose F is analytic on a disk D centered at 0 such that $qD \subseteq D$. If*

$$F(qt^k) = \mu F(t^k) \quad (7)$$

for infinitely many k tending to infinity, then

$$F(z) = \sum_{\substack{n \geq 0 \\ q^n = \mu}} c_n z^n \quad (8)$$

on D . In particular, if $q^n \neq \mu$ for every $n \in \mathbb{N}_0$, then $F \equiv 0$.

Proof. The analytic function

$$H(z) = F(qz) - \mu F(z)$$

has the zeros t^k accumulating at the interior point 0, so $H \equiv 0$. If $F(z) = \sum_{n \geq 0} c_n z^n$, coefficient comparison in $F(qz) = \mu F(z)$ gives $(q^n - \mu)c_n = 0$ for every n . $\square$

Corollary 4.5 (No analytic adjacent strip). *In the oriented Alaoglu–Erdős setting, no nonzero analytic germ at 0 can satisfy*

$$F(Q^k) = R^k, \quad F(\alpha Q^k) = \lambda R^k \quad (9)$$

for every sufficiently large k .

Proof. Apply Theorem 4.4 with $(t, q, \mu) = (Q, \alpha, \lambda)$. There is no $n \in \mathbb{N}_0$ with $2^n = 3$ and no $n \in \mathbb{N}_0$ with $3^n = 2$, so $F \equiv 0$, contradicting $F(Q^k) = R^k \neq 0$. $\square$

Remark 4.6. The edge-law theorem is strictly more flexible than Theorem 4.1: it uses only the ratio between adjacent rows and does not use the formula R^k . Conversely, the one-ray theorem gives a contradiction before the adjacent row is introduced. The two versions are kept because they lead to different finite and functional-analytic targets later.

4.3 Continuity, meromorphic germs, and finite ramification

Analyticity is genuinely stronger than continuity. The support pattern of Q and R controls the first regularity wall.

Proposition 4.7 (Continuity stratification). *Let $p \mid Q$.*

- (i) *If $p \nmid R$, no function continuous at 0 can satisfy $F(Q^k) = R^k$ for all large k .*
- (ii) *If $p \mid R$, the assignment $F(0) = 0$, $F(Q^k) = R^k$ is continuous on the closed subset $\{0, Q^k : k \geq 0\}$, but by Theorem 4.2 it has no analytic extension in the Alaoglu–Erdős setting.*

Proof. If $p \nmid R$ and R^k converged in $\mathbb{Q}_p$, then its limit would be nonzero and the same successive-difference argument as in Theorem 4.1 would force $R = 1$, impossible. If $p \mid R$, then both Q^k and R^k tend to zero, so the displayed assignment is continuous on that subset. $\square$

This gives a useful local taxonomy. An external prime lying in the symmetric difference of $\text{supp}(M)$ and $\text{supp}(A)$ creates a *continuity wall* in one orientation. An external prime common to both supports allows continuity but still creates an *analyticity wall*. The common-support case is therefore not locally invisible; it merely moves the obstruction one rung up the regularity ladder.

Theorem 4.8 (Meromorphic and finite-Puiseux rigidity). *Let K, t, s be as in Theorem 4.1.*

- (i) *If a meromorphic germ F at 0 satisfies $F(t^k) = s^k$ for all sufficiently large k , then $F(z) = z^d$ and $s = t^d$ for some $d \in \mathbb{Z}$.*
- (ii) *If, after a finite ramified cover $z = w^e$, a meromorphic germ realizes those values on a compatible branch, then $s^e = t^d$ for some $d \in \mathbb{Z}$.*

Consequently, the oriented Alaoglu–Erdős ray admits neither a meromorphic germ nor a finite-Puiseux germ at 0.

Proof. For (i), write the Laurent germ as $F(z) = z^d G(z)$ with $d \in \mathbb{Z}$ and $G(0) \neq 0$; the proof of Theorem 4.1 is unchanged. For (ii), work in a complete algebraic closure and choose τ with $\tau^e = t$. Applying (i) to the pullback germ in w gives $s = \tau^d$, hence $s^e = t^d$. In the Alaoglu–Erdős setting, $R^e = Q^d$ would imply $3^e = 2^d$ or $2^e = 3^d$ after raising the defining real identities to the positive power $1/x$, impossible by unique factorization. $\square$

First frontier statement

A hypothetical counterexample cannot be represented at its external accumulation point by any ordinary analytic, meromorphic, or finitely ramified algebraic branch. Therefore every successful interpolation of the conditional grid must be essentially singular at zero, infinitely ramified, nonlocal, or tied to a summation procedure that is not a convergent p -adic germ. The next sections show that the canonical geometric Newton interpolation chooses precisely the last option: an archimedean sum whose p -adic terms explode.

5 The external-prime Newton wall

5.1 The imported geometric Newton coefficients

For integers $Q \geq 2$ and $R \geq 1$, let

$$\phi_n(z) = \prod_{j=0}^{n-1} (z - Q^j), \quad c_n(Q, R) = \frac{\prod_{j=0}^{n-1} (R - Q^j)}{Q^{\binom{n}{2}} \prod_{j=1}^n (Q^j - 1)}, \quad c_0 = 1. \quad (10)$$

The degree- N Newton interpolant is

$$P_N^{Q,R}(z) = \sum_{n=0}^N c_n(Q, R) \phi_n(z). \quad (11)$$

The supplied report proves the exact identity

$$P_N^{Q,R}(Q^k) = R^k \quad (0 \leq k \leq N), \quad (12)$$

and constructs a locally uniformly convergent complex entire completion from the same series. Those facts are imported; the new point here is to inspect the rational coefficients and summands in the completion $\mathbb{Q}_p$ selected by an external prime.

The denominator in (10) is obtained from

$$\prod_{j=0}^{n-1} (Q^n - Q^j) = Q^{\binom{n}{2}} \prod_{h=1}^n (Q^h - 1).$$

When $p \mid Q$, every $Q^h - 1$ is a p -adic unit. Thus the entire local problem is governed by the numerator gaps $R - Q^j$ against the explicit quadratic denominator $Q^{\binom{n}{2}}$.

5.2 Exact valuation of every Newton coefficient

We keep the oriented notation (4): $m = v_p(Q) > 0$ and $a = v_p(R) \geq 0$. Because (Q, R) is an oriented hypothetical Alaoglu–Erdős pair, $R \neq Q^h$ for every $h \in \mathbb{N}_0$; otherwise the defining positive real identities would give $3 = 2^h$ or $2 = 3^h$. Thus all numerator gaps below are nonzero and their valuations are finite.

Theorem 5.1 (Exact external-prime coefficient law). *For $n \geq 1$, the following hold.*

(a) *If $a = 0$, then*

$$v_p(c_n(Q, R)) = v_p(R - 1) - m \binom{n}{2}. \quad (13)$$

(b) *Suppose $a > 0$. Put*

$$q_* = \left\lfloor \frac{a-1}{m} \right\rfloor, \quad \varepsilon = \begin{cases} v_p(R - Q^{a/m}) - a, & m \mid a, \\ 0, & m \nmid a. \end{cases} \quad (14)$$

Then, for every $n \geq q_ + 2$,*

$$v_p(c_n(Q, R)) = m \binom{q_* + 1}{2} + a(n - 1 - q_*) + \varepsilon - m \binom{n}{2}. \quad (15)$$

In particular,

$$v_p(c_n(Q, R)) = -\frac{m}{2}n^2 + \left(a + \frac{m}{2}\right)n + O_{Q,R,p}(1). \quad (16)$$

Proof. Because $p \mid Q$, one has $v_p(Q^h - 1) = 0$ for $h \geq 1$, and hence

$$v_p(c_n) = \sum_{j=0}^{n-1} v_p(R - Q^j) - m \binom{n}{2}. \quad (17)$$

If $a = 0$, the term $j = 0$ is $v_p(R - 1)$, while for every $j \geq 1$ the two terms R and Q^j have distinct valuations 0 and $mj > 0$, so $v_p(R - Q^j) = 0$. This proves (13).

Now let $a > 0$. The $j = 0$ gap $R - 1$ is a unit. For $j \geq 1$ and $mj \neq a$, the ultrametric inequality is strict and gives

$$v_p(R - Q^j) = \min(a, mj).$$

The positive integers j with $mj < a$ are exactly $1, \dots, q_*$. Their contribution is $m \binom{q_* + 1}{2}$. Every later index contributes a , except that if $a = mh$ there is one tie at $j = h$, whose excess above a is exactly ε . The range $n \geq q_* + 2$ includes the first index at or above the tie threshold. Substitution into (17) gives (15); expansion of the binomial coefficient gives (16). $\square$

Remark 5.2 (The tie correction is finite but indispensable). The term ε records the case in which the valuation ray of R meets the valuation ray of Q^j . It can be arbitrarily large for unrelated integer pairs. In a hypothetical Alaoglu–Erdős pair the equality $R = Q^{a/m}$ is impossible, so ε is always finite, but it cannot simply be discarded from exact finite statements.

Corollary 5.3 (Quadratic denominator escape). *Let D_n be the least positive integer such that $D_n c_n(Q, R) \in \mathbb{Z}_{(p)}$, where $\mathbb{Z}_{(p)}$ is the localization away from p . Then*

$$v_p(D_n) = \frac{m}{2}n^2 - \left(a + \frac{m}{2}\right)n + O_{Q,R,p}(1). \quad (18)$$

In particular, the p -part of the denominator grows quadratically in n .

5.3 The quadratic loss is intrinsic, not a Newton-basis artifact

A change of interpolation basis cannot remove the local denominator. The following theorem applies to every polynomial that realizes the same finite data, regardless of its degree.

For a polynomial $P(z) = \sum b_j z^j \in \mathbb{Q}_p[z]$, write

$$\|P\|_G = \max_j |b_j|_p$$

for the unit-disk Gauss norm.

Theorem 5.4 (Universal Gauss-norm lower bound). *Fix $N \geq 0$. Let $P \in \mathbb{Q}_p[z]$ be any polynomial, of arbitrary degree, satisfying*

$$P(Q^k) = R^k \quad (0 \leq k \leq N). \quad (19)$$

Then

$$\|P\|_G \geq |c_N(Q, R)|_p. \quad (20)$$

Consequently, every rational polynomial satisfying (19) has at least one coefficient whose reduced denominator has p -adic exponent at least $\max\{0, -v_p(c_N)\}$.

Proof. Let

$$\Phi_{N+1}(z) = \prod_{k=0}^N (z - Q^k) \in \mathbb{Z}_p[z].$$

It is monic. Divide P by Φ_{N+1} in $\mathbb{Q}_p[z]$:

$$P = S\Phi_{N+1} + T, \quad \deg T \leq N.$$

Because the divisor is monic with coefficients in $\mathbb{Z}_p$, the ordinary long-division algorithm is nonexpanding for the Gauss norm: at each step one subtracts a multiple whose coefficients have norm no larger than the current leading coefficient. Hence $\|T\|_G \leq \|P\|_G$.

At every node Q^k , the divisor vanishes, so $T(Q^k) = P(Q^k) = R^k$. By uniqueness of interpolation in degree at most N , $T = P_N^{Q,R}$. Its leading coefficient is precisely $c_N(Q, R)$, because ϕ_N is monic and all preceding Newton terms have smaller degree. Thus

$$|c_N|_p \leq \|T\|_G \leq \|P\|_G.$$

□

Corollary 5.5 (No bounded exact polynomial family). *For any sequence of rational polynomials P_N satisfying $P_N(Q^k) = R^k$ for $0 \leq k \leq N$,*

$$\log_p \|P_N\|_G \geq \frac{m}{2} N^2 - O(N).$$

Thus exact interpolation through longer native prefixes forces quadratic coefficient escape in every polynomial model, not only in the canonical Newton one.

Remark 5.6. This lower bound is control-sensitive only through the choice of an external prime. Similar blow-up can occur at 2 or 3 for integer controls, but a prime $p \geq 5$ contracting one of the conditional outputs exists only in the nonintegral branch. Any finishing argument still has to use the fact that $p \nmid 6$, not merely the existence of a contracting prime.

5.4 Exact valuation off the native ray

Let $u \in \mathbb{Z}_p^\times$ and $k \in \mathbb{N}_0$. The point uQ^k lies on the same valuation circle as the native node Q^k but is off the ray when $u \neq 1$.

Lemma 5.7 (Off-ray Newton-basis valuation). *For $n > k$,*

$$v_p(\phi_n(uQ^k)) = m \left(kn - \binom{k+1}{2} \right) + v_p(u-1). \quad (21)$$

Proof. In

$$\phi_n(uQ^k) = \prod_{j=0}^{n-1} (uQ^k - Q^j),$$

the factor $j = 0$ is a unit when $k \geq 1$; the formula also covers $k = 0$ directly. For $1 \leq j < k$, the valuation is mj ; at $j = k$ it is $mk + v_p(u-1)$; and for $j > k$ it is mk . Summing gives

$$m \sum_{j=1}^{k-1} j + mk + v_p(u-1) + mk(n-k-1) = m \left(kn - \frac{k(k+1)}{2} \right) + v_p(u-1).$$

□

Put

$$T_n(z) = c_n(Q, R) \phi_n(z). \quad (22)$$

Combining Theorems 5.1 and 5.7 immediately gives:

Corollary 5.8 (Quadratic off-ray explosion). *For fixed $u \in \mathbb{Z}_p^\times$, $u \neq 1$, and fixed k , one has*

$$v_p(T_n(uQ^k)) = -\frac{m}{2} n^2 + \left(a + m \left(k + \frac{1}{2} \right) \right) n + O_{Q,R,p,u,k}(1). \quad (23)$$

Hence $T_n(uQ^k)$ does not tend to zero in $\mathbb{Q}_p$; its p -adic absolute value tends to infinity at a quadratic-exponential rate.

For the adjacent Alaoglu–Erdős row, $u = \alpha \in \{2, 3\}$ and $p \geq 5$, so $v_p(u-1) = 0$. The formula is then especially clean.

5.5 The exact term ratio and last-term dominance

The two quadratic formulas above can be compressed into one exact first-order identity.

Proposition 5.9 (Exact Newton-term ratio). *For every $n \geq 0$ for which $T_n(z) \neq 0$,*

$$\frac{T_{n+1}(z)}{T_n(z)} = \frac{(R - Q^n)(z - Q^n)}{Q^n(Q^{n+1} - 1)}. \quad (24)$$

For $z = uQ^k$, once $n > k$ and $mn > a$,

$$v_p \left(\frac{T_{n+1}(uQ^k)}{T_n(uQ^k)} \right) = a + mk - mn. \quad (25)$$

For fixed complex z in the ordinary absolute value,

$$\frac{T_{n+1}(z)}{T_n(z)} \longrightarrow \frac{1}{Q}. \quad (26)$$

Proof. The quotient c_{n+1}/c_n is

$$\frac{R - Q^n}{Q^n(Q^{n+1} - 1)},$$

and $\phi_{n+1}/\phi_n = z - Q^n$, giving (24). At the external prime, for $n > k$ one has $v_p(uQ^k - Q^n) = mk$, while $mn > a$ gives $v_p(R - Q^n) = a$; $Q^{n+1} - 1$ is a unit. This proves (25). In the ordinary absolute value, both numerator factors are asymptotic to $-Q^n$, whereas the denominator is asymptotic to Q^{2n+1} . $\square$

Equation (26) is the local reason for the complex entire convergence proved in the companion report. Equation (25) points in exactly the opposite direction: its right side tends linearly to $-\infty$.

Theorem 5.10 (Eventual last-term domination). *Fix $u \in \mathbb{Z}_p^\times$, $u \neq 1$, and $k \in \mathbb{N}_0$. Then there exists N_0 such that for all $N \geq N_0$,*

$$v_p(P_N^{Q,R}(uQ^k)) = v_p(T_N(uQ^k)). \quad (27)$$

In particular,

$$v_p(P_N^{Q,R}(uQ^k)) = -\frac{m}{2}N^2 + \left(a + m\left(k + \frac{1}{2}\right)\right)N + O(1). \quad (28)$$

Proof. By (25), the valuations of successive terms are strictly decreasing once $n > k + a/m$. By (23), they tend to $-\infty$. Therefore, for all sufficiently large N , the final term T_N has valuation strictly smaller than every preceding term. In a nonarchimedean field a sum with a unique term of least valuation has the same valuation as that term, proving (27); the asymptotic follows from Theorem 5.8. $\square$

Corollary 5.11 (Adjacent-row repulsion). *In the oriented Alaoglu–Erdős setting, fix $k \in \mathbb{N}_0$ and put $z = \alpha Q^k$. Then, for all sufficiently large N ,*

$$v_p(P_N^{Q,R}(z) - \lambda R^k) = v_p(P_N^{Q,R}(z)) \quad (29)$$

$$= -\frac{m}{2}N^2 + \left(a + m\left(k + \frac{1}{2}\right)\right)N + O(1). \quad (30)$$

Thus the native-ray interpolants become not merely inaccurate but quadratically p -adically repelled from the prescribed adjacent-row integer.

Proof. The target λR^k has fixed finite valuation ka , because λ is a p -adic unit. The valuation in (28) tends to $-\infty$, so eventually it is strictly smaller than ka . The ultrametric inequality then gives (29). $\square$

The p -adic Newton wall

The same rational Newton series exhibits three mutually compatible behaviors.

- (i) At a native node $z = Q^k$, every term with $n > k$ vanishes identically, so the value R^k is a finite algebraic identity valid in every completion.
- (ii) At a fixed complex point, the term ratio tends to $1/Q$, so the series converges geometrically and defines the entire function constructed in the supplied report.
- (iii) At every off-ray point $z = uQ^k$ with u a p -adic unit different from 1, the term valuations tend to $-\infty$ quadratically, and the partial sums are eventually dominated by their last term.

The native ray is therefore a set of exact truncation points surrounded, in the external p -adic geometry, by maximal divergence. This is not a numerical accident and not a choice of summation order; it follows from the exact ratio (24).

5.6 Why the wall is not yet a contradiction

The wall says that the canonical *archimedean* completion of the native interpolation data has no corresponding p -adic convergence off the ray. It does not say that the complex value at an off-ray rational point must equal a p -adic sum of the same terms. An infinite sum of rational numbers may converge in one completion and diverge in another; even when it converges in both, the two limits need not be the same rational number.

At native nodes, equality is place-independent only because the series truncates. At the adjacent row it does not truncate, and Theorem 5.11 shows exactly how the place dependence enters. Any proof that transfers the complex entire completion into a p -adic analytic germ without an additional arithmetic compatibility theorem would be circular.

The product formula also does not close the gap. A nonzero rational residual can have a p -denominator of size $p^{\Theta(N^2)}$ while being only geometrically small in the ordinary absolute value. The elementary lower bound supplied by its denominator is then of order $\exp(-\Theta(N^2))$, far below an archimedean error of order $\exp(-\Theta(N))$. Quadratic denominator escape therefore defeats, rather than helps, a naive unit-locking argument.

Closed shortcut: “the entire function has rational data, so evaluate it p -adically”

The Taylor coefficients of the complex completion are not obtained by a coefficientwise finite rational operation; each fixed Taylor coefficient receives infinitely many Newton contributions. The complex convergence that defines those coefficients is itself place-dependent. Rational values at the native nodes come from exact truncation and do not supply an adelic Taylor expansion. This shortcut cannot be repaired without a separate overconvergence or algebraicity theorem.

6 The Tate-algebra edge operator and finite-strip defects

The analytic contradiction in [Theorem 4.4](#) is qualitative. On finite polynomials the same mechanism is diagonal and quantitative.

6.1 The edge spectrum

For a radius $\rho > 0$ and a polynomial $P(z) = \sum_{j=0}^d b_j z^j \in \mathbb{Q}_p[z]$, put

$$\|P\|_\rho = \max_{0 \leq j \leq d} |b_j|_p \rho^j. \quad (31)$$

Define the edge operator

$$\mathcal{L}_{\alpha, \lambda} P(z) = P(\alpha z) - \lambda P(z). \quad (32)$$

Since $p \geq 5$, $|\alpha|_p = |\lambda|_p = 1$, and

$$\mathcal{L}_{\alpha, \lambda} P = \sum_{j=0}^d (\alpha^j - \lambda) b_j z^j. \quad (33)$$

Definition 6.1 (Finite edge gap). For $d \in \mathbb{N}_0$, set

$$\sigma_{p,d}(\alpha, \lambda) = \min_{0 \leq j \leq d} |\alpha^j - \lambda|_p. \quad (34)$$

It is positive because no ordinary integer power of 2 equals 3, and no ordinary integer power of 3 equals 2.

Proposition 6.2 (Finite spectral inequality). For every $P \in \mathbb{Q}_p[z]$ of degree at most d ,

$$\|\mathcal{L}_{\alpha, \lambda} P\|_\rho \geq \sigma_{p,d}(\alpha, \lambda) \|P\|_\rho. \quad (35)$$

Proof. This is coefficientwise from (33):

$$\max_j |\alpha^j - \lambda|_p |b_j|_p \rho^j \geq \left(\min_j |\alpha^j - \lambda|_p \right) \max_j |b_j|_p \rho^j.$$

□

For the full Tate algebra, define

$$\sigma_p(\alpha, \lambda) = \inf_{j \geq 0} |\alpha^j - \lambda|_p. \quad (36)$$

The closure of the cyclic semigroup $\{\alpha^j : j \geq 0\}$ is compact in $\mathbb{Z}_p^\times$, so

$$\sigma_p(\alpha, \lambda) > 0 \iff \lambda \notin \overline{\{\alpha^j : j \geq 0\}}. \quad (37)$$

If λ is not even a power of α modulo p , then every $\alpha^j - \lambda$ is a p -adic unit and $\sigma_p = 1$.

Proposition 6.3 (Tate spectral gap). If $\sigma_p(\alpha, \lambda) > 0$, then on every Tate algebra on a disk stable under $z \mapsto \alpha z$,

$$\|\mathcal{L}_{\alpha, \lambda} F\|_\rho \geq \sigma_p(\alpha, \lambda) \|F\|_\rho. \quad (38)$$

The operator is injective and has a bounded coefficientwise inverse on its image.

Proof. The proof of [Theorem 6.2](#) passes to convergent series. Injectivity also follows from the absence of an exact eigenvalue $\alpha^j = \lambda$. $\square$

Remark 6.4 (Resonant primes). When λ lies in the p -adic closure of powers of α , the infinite gap vanishes, but every finite gap (34) remains positive. The quantities

$$\mu_{p,d}(\alpha, \lambda) := -\log_p \sigma_{p,d}(\alpha, \lambda) = \max_{0 \leq j \leq d} v_p(\alpha^j - \lambda)$$

measure the exact small-divisor loss. This separates two local branches: *nonresonant* external primes, for which the edge operator is uniformly bounded below, and *resonant* external primes, where increasingly accurate congruences $\alpha^j \equiv \lambda \pmod{p^r}$ may occur. Any argument using a Tate inverse must keep this branch distinction explicit.

6.2 A finite-strip quotient lower bound

Let $K \leq N$ and define the tail-node polynomial

$$\Phi_{K,N}(z) = \prod_{k=K}^N (z - Q^k). \quad (39)$$

If a polynomial P satisfies the adjacent-row identity at those nodes, then $\mathcal{L}_{\alpha,\lambda}P$ is divisible by $\Phi_{K,N}$.

Theorem 6.5 (Finite-strip defect theorem). *Let $P \in \mathbb{Q}_p[z]$ have degree at most d and suppose*

$$P(\alpha Q^k) = \lambda P(Q^k) \quad (K \leq k \leq N). \quad (40)$$

Write

$$\mathcal{L}_{\alpha,\lambda}P(z) = \Phi_{K,N}(z) S(z). \quad (41)$$

Fix $0 < \rho < 1$ such that $|Q|_p^K \leq \rho$. If additionally $P(Q^K) = R^K$, then

$$\|S\|_\rho \geq \sigma_{p,d}(\alpha, \lambda) |R|_p^K \rho^{-(N-K+1)}. \quad (42)$$

Proof. The roots in (40) give the factorization (41). The weighted Gauss norm is multiplicative, and because $|Q|_p^k \leq |Q|_p^K \leq \rho$ for every $k \geq K$,

$$\|\Phi_{K,N}\|_\rho = \prod_{k=K}^N \max(\rho, |Q|_p^k) = \rho^{N-K+1}. \quad (43)$$

Evaluation inside the disk gives

$$|R|_p^K = |P(Q^K)|_p \leq \|P\|_\rho.$$

Apply [Theorem 6.2](#) and multiplicativity:

$$\rho^{N-K+1} \|S\|_\rho = \|\mathcal{L}_{\alpha,\lambda}P\|_\rho \geq \sigma_{p,d} \|P\|_\rho \geq \sigma_{p,d} |R|_p^K.$$

$\square$

Corollary 6.6 (No tame defect quotient). *Let P_N be the unique polynomial of degree at most $2N + 1$ interpolating both rows*

$$P_N(Q^k) = R^k, \quad P_N(\alpha Q^k) = \lambda R^k \quad (0 \leq k \leq N).$$

Fix K with $|Q|_p^K \leq \rho < 1$. In the factorization

$$\mathcal{L}_{\alpha, \lambda} P_N = \Phi_{K, N} S_{K, N},$$

one has

$$\|S_{K, N}\|_\rho \geq \sigma_{p, 2N+1}(\alpha, \lambda) |R|_p^K \rho^{-(N-K+1)}. \quad (44)$$

At a nonresonant external prime, the quotient norm therefore grows at least geometrically like ρ^{-N} .

The zeros forced by a long adjacent strip make the defect small on a strict disk, but the edge spectral inequality prevents the whole polynomial from becoming small. The quotient $S_{K, N}$ must absorb the difference. This is the finite analogue of the essential singularity forced by [Theorem 4.5](#).

Target 6.7 (Subexponential defect quotient). *Find an arithmetic construction of exact finite two-row interpolants for which, at some external prime and some fixed $\rho < 1$,*

$$\log \|S_{K, N}\|_\rho = o(N) \quad \text{and} \quad -\log \sigma_{p, \deg P_N}(\alpha, \lambda) = o(N).$$

Then (42) is impossible for large N , proving the conjecture in that branch.

The first requirement is the substantive one. Canonical polynomial interpolation generally has much larger local height, and [Theorem 5.4](#) shows that no native-ray exact interpolant can be uniformly bounded. Nevertheless, [Theorem 6.7](#) is a precise finite target: it asks for control of the *edge quotient*, not of every coefficient of the interpolant.

7 Overconvergence, compactness, and a Robba-to-Tate bridge

7.1 A normal-family obstruction

The local classification theorem can be reached from finite data by compactness, provided one has genuine overconvergence. For a convergent series $F(z) = \sum_{j \geq 0} b_j z^j$ on the closed disk of radius ρ , write $\|F\|_\rho = \sup_j |b_j|_p \rho^j$.

Theorem 7.1 (No uniformly overconvergent interpolation family). *Let $\rho > 1$. Suppose $F_N(z) = \sum_{j \geq 0} b_{N, j} z^j$ is analytic on the closed p -adic disk of radius ρ , and suppose*

$$F_N(Q^k) = R^k \quad (0 \leq k \leq N). \quad (45)$$

Then the norms $\|F_N\|_\rho$ are unbounded as $N \rightarrow \infty$. The same conclusion holds if (45) is replaced by the two-row prefix.

Proof. Assume $\|F_N\|_\rho \leq C$. Then $|b_{N, j}|_p \leq C \rho^{-j}$. Closed bounded subsets of $\mathbb{Q}_p$ are compact, so a diagonal subsequence makes every coefficient $b_{N, j}$ converge. The geometric bound on the tails makes that subsequence converge uniformly on the unit disk to an analytic function $F(z) = \sum b_j z^j$. For each fixed k , the equality $F_N(Q^k) = R^k$ holds eventually, so uniform convergence gives $F(Q^k) = R^k$. This contradicts [Theorem 4.2](#). The two-row version is immediate as well. $\square$

Remark 7.2. Boundedness on the unit disk alone is not enough for this compactness argument: coefficient bounds need not give a uniformly small tail. The strict radius gain $\rho > 1$ is exactly the overconvergence input. This is why a Robba-ring or punctured-disk construction is not yet a solution; it must be shown to descend through the puncture with a quantitative radius margin.

Corollary 7.3 (Coefficient escape for any overconvergent model). *Any sequence of exact finite-prefix models analytic on one common disk of radius $\rho > 1$ must have*

$$\limsup_{N \rightarrow \infty} \|F_N\|_\rho = \infty.$$

For polynomial models, Theorem 5.4 gives the stronger explicit lower bound $p^{mN^2/2 - O(N)}$ already on the unit disk.

7.2 The removable-singularity interface

The grid values themselves naturally live on a punctured set converging to zero. It is easy to imagine functions on a punctured disk that realize arbitrary values there while having an essential singularity at zero. The useful theorem would not be another interpolation existence statement, but a descent theorem that makes the singularity removable.

Target 7.4 (Robba-to-Tate descent). *Construct, from the integer grid alone, a function or finite-rank difference module over a Robba ring at an external prime with all three properties:*

- (i) *it realizes the native and adjacent row values in (9);*
- (ii) *its slopes, denominators, or Frobenius structure satisfy an arithmetic boundedness condition derived from $M, A \in \mathbb{Z}$;*
- (iii) *that boundedness forces descent to a Tate algebra, a meromorphic germ, or a finite Puiseux extension at zero.*

Any such descent proves the conjecture by Theorems 4.5 and 4.8.

This target is deliberately modular. It does not demand that the power branch itself be postulated p -adically. It asks for an arithmetic object first and regularity second. The Newton wall says what the descent theorem must defeat: the obvious one-axis completion has quadratically escaping denominators and no off-ray p -adic summation.

7.3 A coefficient-growth criterion

Let $F(z) = \sum_{n \geq n_0} b_n z^n$ be a formal Laurent series over $\mathbb{Q}_p$. It defines a meromorphic germ on a punctured neighborhood of zero whenever the positive tail has positive radius of convergence and the negative tail is finite. By Theorem 4.8, no such series can realize the oriented ray.

Thus any formal object carrying the data must satisfy at least one of:

1. infinitely many negative powers;
2. zero p -adic radius for the positive tail;
3. exponents with unbounded denominators, so that no finite ramified cover turns the object into a Laurent series;

4. failure of the formal values to agree with the integer samples under convergent evaluation.

This four-way escape list is useful when auditing proposed constructions. For example, a G -function-style series with algebraic coefficients, controlled denominators, and positive p -adic radius at almost every prime would immediately be incompatible with an external Alaoglu–Erdős ray, provided its sample equalities were algebraic identities rather than purely archimedean limits.

Target 7.5 (Adelic positive-radius criterion). *Produce a series $F(z) = \sum b_n z^n$ with coefficients in one number field such that:*

- (i) *the equalities $F(Q^k) = R^k$ and $F(\alpha Q^k) = \lambda R^k$ are certified by finite algebraic identities or by a summation theorem valid in every completion;*
- (ii) *at every prime outside a fixed set, and in particular at an external p , the coefficients have at most exponential denominator growth, so the p -adic radius is positive.*

Then the external-prime edge theorem closes the conjecture.

The first clause is essential. Complex equality of an infinite sum to an integer is not an adelic identity.

8 A finite-rank q -difference route

8.1 Why two independent dilations are structurally promising

A genuine interpolation of the conditional grid would carry two dilation laws,

$$F(2z) = 3F(z), \quad F(Mz) = AF(z). \quad (46)$$

Under a nonintegral solution, 2 and M are multiplicatively independent by [Theorem 3.1](#). Systems of linear q -difference equations for two multiplicatively independent dilation parameters are highly rigid under rationality, finite-rank, and consistency hypotheses. Results of Bézivin and Boutabaa, and the later unified treatment of Schäfke and Singer, show that common formal or meromorphic solutions of sufficiently regular independent difference systems often reduce to rational or constant normal forms [8, 9, 10].

The present scalar equations are already consistent: the constants commute and $3A = A3$. Their formal Laurent obstruction is immediate. If $F(z) = \sum_{n \in \mathbb{Z}} c_n z^n$ has only finitely many negative powers, then the first equation in (46) gives

$$(2^n - 3)c_n = 0$$

for every n , hence $F = 0$. On the universal cover of $\mathbb{C}^\times$, however, the nonzero solution z^θ exists. The gap is exactly local regularity and monodromy, not consistency.

8.2 Why existing two- q theorems do not apply automatically

The integer table gives values at a discrete semigroup, not a finite-dimensional difference module over $\overline{\mathbb{Q}}(z)$. The complex geometric Newton completion satisfies a one-axis *inhomogeneous* equation with a q -product defect; it does not satisfy both homogeneous laws in (46). The external-prime

Newton wall further shows that this completion has no off-ray p -adic germ from which one could read a local difference module.

Therefore it would be an overstatement to invoke a two- q rationality theorem directly. One first needs a finite-rank closure of the arithmetic defects.

Target 8.1 (Finite-rank edge cocycle). *Construct a vector $Y(z)$ of finitely many arithmetic interpolation functions and rational matrices $B_2(z), B_M(z) \in \mathrm{GL}_r(\overline{\mathbb{Q}}(z))$ such that*

$$Y(2z) = B_2(z)Y(z), \quad Y(Mz) = B_M(z)Y(z), \quad (47)$$

with the consistency identity

$$B_2(Mz)B_M(z) = B_M(2z)B_2(z), \quad (48)$$

and with one coordinate of Y carrying the exact conditional integer grid. Prove that the system is regular singular or overconvergent at an external prime. A reduction theorem for consistent independent q -difference systems would then place the coordinate in a Laurent/Puiseux class, contradicted by Theorem 4.8.

This target is narrower than “find an analytic interpolation.” Finite rank can sometimes be proved from recurrences among defects even when no scalar completion is regular. The companion report’s exact endpoint defect and two-axis connection formulas provide raw material, but their current defect family contains nonrational infinite q -products and is not known to close in finite rank.

8.3 A practical search protocol

A computational search for Theorem 8.1 should not fit values blindly. It should:

1. choose a small list of canonical one-axis completions and endpoint defects from the supplied report;
2. apply both dilation operators repeatedly and reduce every expression in a fixed symbolic basis of q -products and Newton kernels;
3. compute the rank of the span over $\mathbb{Q}(z)$ at increasing truncation order;
4. reject bases whose rank grows with the truncation order;
5. for any apparently stable rank, solve the exact rational-function consistency equations and verify them symbolically;
6. only then analyze local slopes at primes $p \mid MA$, with external primes separated from 2 and 3.

A stable finite rank would be a substantive new structure theorem. Failure with rank growth would close another route without pretending that numerical near-dependence is a proof.

9 Adelic and control audits

9.1 The integer controls remain an important trap

For a control exponent $x = d \in \mathbb{N}$, the real branch z^θ still has no 2-adic analytic germ interpolating $(2^{dk}, 3^{dk})$. Therefore the mere statement “the irrational power branch is not p -adically analytic”

is control-blind. The external-prime program becomes potentially discriminating only when a construction uses $p \nmid 6$ to obtain regularity, bounded denominators, an edge spectral gap, or a finite-rank local module. Any proposed bridge should answer the concrete question:

Which step works at a prime $p \geq 5$ dividing a hypothetical output but has no analogue at the structural primes 2 and 3 of the integer controls?

The exact valuation formulas provide one possible answer: at an external prime, the quadratic denominator lies outside $\mathbb{Z}[1/6]$ and directly interacts with an edge operator whose two multipliers are units. At present that interaction diagnoses singularity; it does not yet reverse it into regularity.

9.2 No contradiction from the product formula at current scales

The canonical off-ray term has

$$\log |T_n(uQ^k)|_p = \frac{m}{2}n^2 \log p + O(n),$$

while in the ordinary absolute value its ratio tends to $1/Q$, so $\log |T_n|_\infty = -n \log Q + O(1)$ at fixed z away from accidental zeros. These two scales do not oppose one another: the remaining finite places absorb the quadratic balance. A useful product-formula object would need either

1. quadratic archimedean smallness matching the external denominator;
2. cancellation of the external denominator in the *reduced* rational residual; or
3. simultaneous positive valuation at enough other places to exceed the global height.

None follows from the one-axis Newton formula.

9.3 Sampling is not regularity

The points Q^k accumulate at zero p -adically, but a function specified only on those points need not be continuous, let alone analytic. Integrality of the values does not change that. [Theorems 4.1](#) and [4.4](#) should therefore be read as *closure theorems*: once a regular germ exists, the conjecture closes immediately. The construction of that germ, or of a finite-rank object forced to become one, is the missing arithmetic theorem.

Logical boundary

No argument in this report infers p -adic continuity from real continuity, infers a p -adic sum from a complex sum, or treats integer-valued sampling as an analytic extension theorem. Those three inferences are precisely where several attractive one-page “proofs” fail.

10 Exact computational audit

The new valuation statements are symbolic, but an exact audit is useful for detecting index errors, especially in the valuation-tie case. The audit uses Python’s `fractions.Fraction`; no floating-point arithmetic is involved. For each test case it constructs c_n , $\phi_n(uQ^k)$, the Newton term T_n , and

the partial sum P_N as exact rational numbers, then compares directly computed valuations with the formulas in [Theorems 5.1 and 5.7](#).

The test script used for this report has SHA-256 digest

20458cc7e3174c8a7d30f009e8259eba1f6629b044fdc33a997b495a6bdd49fe.

Its complete source is reproduced in [Section A](#).

10.1 Case I: unit output at the contracting prime

Take

$$Q = 35, \quad R = 22, \quad p = 5, \quad u = 2, \quad k = 2.$$

Here $m = 1$, $a = 0$, and $v_p(R - 1) = 0$. Thus

$$v_p(c_n) = -\binom{n}{2}, \quad v_p(\phi_n(2Q^2)) = 2n - 3 \quad (n > 2),$$

so

$$v_p(T_n(2Q^2)) = -\frac{(n-2)(n-3)}{2} \quad (n > 2). \quad (49)$$

The exact audit is:

Table 1: Case I: exact valuations.

n	$v_p(c_n)$	$v_p(\phi_n)$	$v_p(T_n)$	$v_p(P_n)$
0	0	0	0	0
1	0	0	0	0
2	-1	1	0	0
3	-3	3	0	1
4	-6	5	-1	-1
5	-10	7	-3	-3
6	-15	9	-6	-6
7	-21	11	-10	-10
8	-28	13	-15	-15
9	-36	15	-21	-21
10	-45	17	-28	-28
11	-55	19	-36	-36
12	-66	21	-45	-45

From $n = 4$ onward, the final term is already the unique term of least valuation, so the partial sum has exactly the predicted last-term valuation.

10.2 Case II: a genuine valuation tie

Take

$$Q = 10, \quad R = 110, \quad p = 5, \quad u = 2, \quad k = 1.$$

Now $m = a = 1$, and the tie occurs at $j = 1$:

$$\varepsilon = v_p(110 - 10) - 1 = v_p(100) - 1 = 1.$$

Formula (15) gives, for $n \geq 2$,

$$v_p(c_n) = n - \binom{n}{2}.$$

The audit confirms that omitting ε would shift every coefficient valuation from $n = 2$ onward.

Table 2: Case II: exact valuations with a nonzero tie correction.

n	$v_p(c_n)$	$v_p(\phi_n)$	$v_p(T_n)$	$v_p(P_n)$
0	0	0	0	0
1	0	0	0	1
2	1	1	2	1
3	0	2	2	1
4	-2	3	1	1
5	-5	4	-1	-1
6	-9	5	-4	-4
7	-14	6	-8	-8
8	-20	7	-13	-13
9	-27	8	-19	-19
10	-35	9	-26	-26
11	-44	10	-34	-34
12	-54	11	-43	-43

10.3 Case III: integral valuation slope but unit mismatch

Take

$$Q = 25, \quad R = 75, \quad p = 5, \quad u = 2, \quad k = 1.$$

Here $m = a = 2$, so $a/m = 1$ is an integral candidate order. Nevertheless $R/Q = 3 \neq 1$, exactly the unit obstruction in [Theorem 4.3](#). The tie excess is zero, and

$$v_p(c_n) = -(n-1)(n-2) \quad (n \geq 2).$$

Table 3: Case III: an integral slope does not prevent quadratic escape.

n	$v_p(c_n)$	$v_p(\phi_n)$	$v_p(T_n)$	$v_p(P_n)$
0	0	0	0	0
1	0	0	0	2
2	0	2	2	2
3	-2	4	2	2
4	-6	6	0	0
5	-12	8	-4	-4
6	-20	10	-10	-10
7	-30	12	-18	-18
8	-42	14	-28	-28
9	-56	16	-40	-40
10	-72	18	-54	-54
11	-90	20	-70	-70
12	-110	22	-88	-88

All formula checks through $n = 12$ pass exactly in the three branches. The cases are not intended as candidate Alaoglu–Erdős outputs; they are local unit tests covering $a = 0$, a positive valuation with nonzero tie excess, and an integral slope with unit mismatch.

10.4 What to compute next

The next computational experiment should not scan arbitrary integer pairs for a false counterexample. It should test the finite-rank and defect-quotient targets:

1. For many pairs (Q, R) and external primes $p \mid Q$, construct the exact two-row interpolant through $Q^k, \alpha Q^k$ for $0 \leq k \leq N$.
2. Factor $\mathcal{L}_{\alpha, \lambda} P_N$ by the largest forced tail product $\Phi_{K, N}$ and record the weighted Gauss norm of the quotient.
3. Separate primes by the resonance statistic $\mu_{p, d}(\alpha, \lambda)$.
4. Compare actual pairs arising from the supplied finite search with integer controls and with random pairs having the same valuation profile.
5. Apply repeated 2- and Q -dilations to the canonical defect family and compute exact ranks over $\mathbb{Q}(z)$, looking for stabilization rather than numerical singular values.

A stable finite rank or subexponential defect quotient would be a positive discovery. Growth linear in N for the rank, or geometric growth saturating (44), would be a clean negative closure.

11 Lean formalization plan

The new package splits naturally into finite algebra, elementary valuation combinatorics, and nonarchimedean analysis. The first two layers are substantially easier to formalize than the full analytic identity theorem and should be completed first.

11.1 Proposed module graph

Table 4: Proposed Lean modules and trust boundary.

Module	Initial status	Main content
ExternalPrimeOrientation	finite / elementary	The external-prime lemma; orientation data; unit facts for $p \geq 5$; impossibility of $R = \mathbb{Q}^d$ in either Alaoglu–Erdős orientation.
QNewtonCoeffValuation	Lean-ready	Formula (17); $a = 0$ branch; threshold q_* ; unique tie and excess ε ; exact formula (15).
OffRayNewtonBasis	Lean-ready	Factor-by-factor proof of (21); exact term ratio (24); eventual negative ratio valuation.
NewtonLastTermDominance	Lean-ready	Strictly decreasing valuations, unique-minimum valuation of a finite sum, and (28).
InterpolationGaussLowerBound	finite polynomial	Monic division over $\mathbb{Q}_p$; nonexpansion of Gauss norm; uniqueness of the interpolation remainder; Theorem 5.4 .
EdgeOperatorSpectrum	finite polynomial	Diagonal coefficient action, finite gap $\sigma_{p,d}$, and inequality (35).
FiniteStripDefect	finite polynomial	Root factorization, weighted Gauss norm of $\Phi_{K,N}$, and Theorem 6.5 .
GeometricRayAnalytic	analysis layer	Order of an analytic germ, convergence of powers, the nonarchimedean identity theorem, and Theorem 4.1 .
EdgeLawAnalytic	analysis layer	Strassmann/identity theorem for accumulating zeros and coefficient-spectrum theorem Theorem 4.4 .
OverconvergentCompactness	analysis/topology	Diagonal compactness of bounded coefficient balls, uniform-tail convergence, and Theorem 7.1 .

11.2 Finite theorem signatures

The valuation layer can be stated without committing to a particular analytic API. A possible pseudocode interface is:

```
-- Q, R, p, n are natural numbers; p is prime and p | Q.
def qNewtonCoeff (Q R n : N) : Q :=
  (prod j in range n, (R : Z) - Q^j) /
  (Q^(n.choose 2) * prod j in Icc 1 n, (Q^j : Z) - 1)

theorem padicVal_qNewtonCoeff_of_val_eq_zero
  (hpQ : p | Q) (hR : padicValNat p R = 0) (hn : 1 <= n) :
  padicValRat p (qNewtonCoeff Q R n) =
  padicValInt p (R - 1) - padicValNat p Q * n.choose 2

theorem padicVal_qNewtonCoeff_eventually
  (hpQ : p | Q) (ha : 0 < padicValNat p R)
  (hn : qStar Q R p + 2 <= n) :
  padicValRat p (qNewtonCoeff Q R n) =
  exactLocalFormula Q R p n

theorem padicVal_qNewtonTerm_offRay
```

```

    (hpQ : p | Q) (hpu : not p | u) (hk : k < n) :
    padicValRat p
    (qNewtonCoeff Q R n * prod j in range n, (u*Q^k : Z) - Q^j) =
    exactTermFormula Q R p u k n

```

The actual Mathlib valuation types may favor additive valuations into $\mathbb{Z}_{m\infty}$ rather than the schematic integer-valued functions above. The important design choice is to isolate all infinity cases ($R - Q^j = 0$) at the boundary. For an Alaoglu–Erdős orientation those equalities are impossible, but the generic library theorem should either assume them away or state the valuation in an extended codomain.

11.3 Recommended proof order

1. Formalize the denominator valuation $v_p(Q^{\binom{n}{2}} \prod_{j=1}^n (Q^j - 1)) = m^{\binom{n}{2}}$.
2. Formalize the finite minimum sum $\sum_{j=1}^{n-1} \min(a, mj)$ with the threshold q_* . Keep the tie correction as a separate lemma rather than burying it in case splits.
3. Prove the off-ray factor valuations by partitioning the index range into $j < k$, $j = k$, and $j > k$.
4. Prove the exact term ratio algebraically before any asymptotics.
5. Derive eventual strict decrease using ordinary integer inequalities; then use the unique-minimum valuation lemma for finite sums.
6. Formalize monic-division nonexpansion and the universal Gauss bound.
7. Add the edge-operator finite spectrum and finite-strip factorization.
8. Only after the finite package is kernel-accepted, connect to Mathlib’s nonarchimedean analytic-function and power-series APIs for the identity theorem.

This order produces useful machine-checked artifacts even if the analytic library work takes longer. In particular, [Theorems 5.1](#), [5.4](#) and [5.10](#) are finite statements with no reliance on compactness, choice of branch, or complex analysis.

11.4 Formal status that should be claimed

Until code is compiled by Lean with no `sorry`, the report should retain the following status:

Statement	Current status
External-prime analytic rigidity	proved here, paper level
Exact coefficient valuation	proved here, exact Python audit
Universal Gauss lower bound	proved here, paper level
Off-ray valuation and last-term dominance	proved here, exact Python audit
Finite edge spectrum and strip defect	proved here, paper level
Robba-to-Tate and finite-rank interfaces	conditional research targets
Full Alaoglu–Erdős conjecture	open

12 Priority research program

12.1 Priority 1: formalize the finite Newton wall

The finite local package is exact, short enough for independent checking, and directly usable in later experiments. Formalization should include the resonant tie correction and last-term dominance, not only the quadratic asymptotic. This is the fastest route to a kernel-certified contribution distinct from the supplied report.

12.2 Priority 2: measure the edge-defect quotient

For exact two-row interpolation polynomials, compute

$$S_{K,N}(z) = \frac{P_N(\alpha z) - \lambda P_N(z)}{\prod_{k=K}^N (z - Q^k)}$$

at external primes. The decisive statistic is

$$\frac{1}{N} \log \|S_{K,N}\|_\rho + \log \rho - \frac{1}{N} \log \sigma_{p, \deg P_N}.$$

A limit bounded away from zero confirms saturation of the finite-strip lower bound. A limit at or below zero would signal exactly the subexponential quotient needed by [Theorem 6.7](#) and should be investigated symbolically.

12.3 Priority 3: search for finite-rank closure of defects

The supplied report contains several exact one-axis and two-axis defect identities. Treat them as generators of a module under the two dilation operators, not as isolated special functions. Rank stabilization over $\mathbb{Q}(z)$ is the positive signal. Every candidate relation must be reconstructed exactly and then verified by symbolic polynomial identities.

12.4 Priority 4: classify external-prime resonance

For a candidate pair (M, A) , label every external prime by:

$$p \mid M : \quad 3 \in \overline{\langle 2 \rangle} \subset \mathbb{Z}_p^\times? \quad p \mid A : \quad 2 \in \overline{\langle 3 \rangle} \subset \mathbb{Z}_p^\times?$$

At nonresonant primes the Tate edge operator has a uniform inverse bound. At resonant primes, record $\mu_{p,d}$ and compare it with the degree and defect growth. A theorem forcing at least one nonresonant external orientation would materially strengthen the route; no such theorem is currently known.

12.5 Priority 5: look for an arithmetic overconvergence theorem

The best possible bridge would start from a globally defined arithmetic object: a diagonal of a rational function, a G -function, a finite-rank q -difference module, or another class with controlled denominators at almost every prime. If the exact integer samples were certified within that class, an external prime would force positive radius and the local rigidity theorem would finish. The construction must be genuinely algebraic across places; an archimedean interpolation with rational sample values is not enough.

12.6 Explicit kill criteria

To avoid reopening closed avenues, each program has a finite kill criterion:

Table 5: Positive signals and kill criteria.

Program	Positive signal	Kill criterion
Defect quotient	$\log \ S_{K,N}\ _\rho = o(N)$ after the finite spectral loss	exact growth $\geq (c+o(1))N$ with $c > -\log \rho$ in every admissible construction
Finite-rank cocycle	rank over $\mathbb{Q}(z)$ stabilizes	exact rank grows unboundedly with truncation order
External resonance	one external orientation has $\sigma_p > 0$	all external orientations are resonant, with small divisors matching every gain
Overconvergence	one number-field series has positive radius and place-independent samples	coefficient denominators have quadratic escape at every external orientation
Product formula	reduced residual has matching quadratic archimedean decay	reduced height remains quadratic while decay is only linear

13 Honest bottom line

No proof of [Theorem 1.1](#) has been obtained in this report. The Four Exponentials barrier has not been bypassed, and the rival team’s rumored Lean development could not be verified from public sources.

There is nevertheless a concrete advance relative to the supplied corpus. A hypothetical counterexample has an external nonarchimedean accumulation point, and at that point:

1. one geometric ray already excludes every analytic or meromorphic germ unless the two output integers are exact integral powers of one another;
2. one adjacent row excludes every finite-Puiseux germ by a coefficient-spectrum argument;
3. every exact finite polynomial interpolation of the native ray incurs an intrinsic quadratic external-prime denominator, independent of basis;
4. the canonical geometric Newton terms diverge quadratically at every off-ray unit multiple, and the partial sums are eventually equal in valuation to their final term;
5. every exact finite adjacent strip forces an exponentially large edge-defect quotient on a strict disk, up to the explicitly isolated small-divisor factor;
6. any uniformly overconvergent family of finite-prefix interpolants would compactify to the forbidden analytic germ.

The remaining problem is no longer “find some interpolation.” Interpolation is abundant. The sharp bridge is:

The new finishing interface

Derive, from the global arithmetic relation $M = 2^x$, $A = 3^x$ and the integer semigroup alone, either

- (i) a place-independent positive-radius interpolation at one external prime;
- (ii) a Robba-ring object forced to descend to a Tate or finite-Puiseux germ;
- (iii) a finite-rank consistent two-dilation module regular at the external accumulation point; or
- (iv) a subexponential finite-strip edge quotient contradicting (42).

Any one of these closes the conjecture immediately. The exact Newton-wall theorems explain why the most obvious completion fails and quantify the denominator growth that a successful construction must cancel.

This is a narrower and more testable frontier than the initial statement. It also suggests a useful interpretation of any genuine rival breakthrough: unless it introduces new transcendence input, it should be possible to locate its decisive step as an adelic regularity theorem, a finite-rank difference closure, or an external-prime denominator cancellation. Those are now explicit interfaces rather than vague hopes.

A Exact audit code

```

from fractions import Fraction
from math import prod

def vp_int(n, p):
    if n == 0:
        return 10**9 # infinity sentinel; excluded in assertions
    n = abs(n)
    e = 0
    while n % p == 0:
        e += 1
        n //= p
    return e

def vp_frac(x, p):
    return vp_int(x.numerator, p) - vp_int(x.denominator, p)

def c_n(Q, R, n):
    if n == 0:
        return Fraction(1)
    num = prod(R - Q**j for j in range(n))
    den = Q**(n*(n-1)//2) * prod(Q**j - 1 for j in range(1, n+1))
    return Fraction(num, den)

def phi(Q, z, n):
    return prod(z - Q**j for j in range(n))

```

```

def predicted_c_val(Q, R, p, n):
    m = vp_int(Q, p)
    a = vp_int(R, p)
    if n == 0:
        return 0
    if a == 0:
        return vp_int(R - 1, p) - m*n*(n-1)//2
    qstar = (a - 1)//m
    epsilon = 0
    if a % m == 0:
        h = a//m
        if n - 1 >= h:
            epsilon = vp_int(R - Q**h, p) - a
    if n < qstar + 2:
        return (sum(vp_int(R - Q**j, p) for j in range(n))
                - m*n*(n-1)//2)
    return (m*qstar*(qstar+1)//2
            + a*(n - 1 - qstar)
            + epsilon
            - m*n*(n-1)//2)

def predicted_phi_val(Q, u, k, p, n):
    m = vp_int(Q, p)
    if n <= k:
        return vp_int(phi(Q, u*Q**k, n), p)
    return m*(k*n - k*(k+1)//2) + vp_int(u - 1, p)

cases = [
    (35, 22, 5, 2, 2, "unit-output case"),
    (10, 110, 5, 2, 1, "resonant valuation tie"),
    (25, 75, 5, 2, 1, "integral slope, unit mismatch"),
]

for Q, R, p, u, k, label in cases:
    partial = Fraction(0)
    for n in range(13):
        coeff = c_n(Q, R, n)
        basis = phi(Q, u*Q**k, n)
        term = coeff*basis
        partial += term
    assert vp_frac(coeff, p) == predicted_c_val(Q, R, p, n)
    assert vp_int(basis, p) == predicted_phi_val(Q, u, k, p, n)
    print(label, n, vp_frac(coeff, p), vp_int(basis, p),
          vp_frac(term, p), vp_frac(partial, p))

```

B Proof details for monic division and weighted Gauss norms

For completeness, this appendix records the two standard norm facts used in [Theorems 5.4](#) and [6.5](#).

Lemma B.1 (Monic division is nonexpanding). *Let K be nonarchimedean, let $G \in O_K[z]$ be*

monic, and divide $P = QG + R$ with $\deg R < \deg G$. Then

$$\|Q\|_G, \|R\|_G \leq \|P\|_G.$$

Proof. Run long division from the highest degree downward. At each step the quotient coefficient is the current leading coefficient, whose norm is at most the current polynomial norm. Since all coefficients of G have norm at most one, subtracting that coefficient times a shift of G cannot increase the norm by the ultrametric inequality. Induction proves the claim. $\square$

Lemma B.2 (Norm of a linear factor). *For the weighted Gauss norm (31),*

$$\|z - a\|_\rho = \max(\rho, |a|).$$

The weighted Gauss norm is multiplicative on polynomials, so

$$\left\| \prod_i (z - a_i) \right\|_\rho = \prod_i \max(\rho, |a_i|).$$

Proof. The first formula is immediate from the two coefficients. Multiplicativity is the Gauss lemma for a nonarchimedean norm, after rescaling the variable when ρ lies in the value group. For arbitrary real $\rho > 0$, extend scalars to a complete algebraic closure, whose value group is dense in $\mathbb{R}_{>0}$, approximate ρ by value-group radii, and use continuity of the finite maximum defining the polynomial norm. Only the polynomial case is used here. $\square$

C Compact theorem inventory

Table 6: Inventory of the new package.

Statement	Status	Mathematical role
Geometric-ray rigidity, Theorem 4.1	proved here	Classifies every analytic germ on a contracting geometric ray.
Accumulating edge law, Theorem 4.4	proved here	Converts adjacent-row data into an exact Taylor-spectrum condition.
Meromorphic/Puiseux exclusion, Theorem 4.8	proved here	Shows finite ramification cannot hide the non-integral exponent.
Exact coefficient law, Theorem 5.1	proved and audited	Gives the full quadratic local denominator, including ties.
Universal Gauss bound, Theorem 5.4	proved here	Makes the denominator loss independent of interpolation basis and degree.
Off-ray valuation, Theorem 5.7	proved and audited	Computes the exact local cost of moving from the native ray.
Term ratio and last-term dominance, Theorems 5.9 and 5.10	proved and audited	Establishes maximal p -adic divergence and adjacent-row repulsion.
Finite edge spectrum, Theorem 6.2	proved here	Isolates the exact small-divisor loss.
Finite-strip defect, Theorem 6.5	proved here	Forces a large quotient behind every long exact adjacent strip.
No overconvergent family, Theorem 7.1	proved here	Turns any uniform radius gain into the forbidden analytic germ.
Robba-to-Tate descent, Theorem 7.4	open target	Most direct local finishing bridge.

Statement	Status	Mathematical role
Finite-rank edge cocycle, Theorem 8.1	open target	Connects the arithmetic grid to independent- q rigidity theory.
Subexponential defect quotient, Theorem 6.7	open target	Finite, computationally testable finishing criterion.

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